MTH6134 Statistical Modelling II

Writing a Statistical Report: Tips and Guidance

This document gives guidance on writing a short statistical report for MTH6134. A good report is not just a correct analysis, it is a well-communicated one. The goal is to present your findings clearly and concisely so that a reader can follow your reasoning without needing to run your code themselves.

Structure

A short statistical report should have the following sections - but always read the bried for the specific assessment for variations and specifications.

1. **Introduction or executive summary.** State the research question and the aim of the analysis. A reader should understand what the report is about from this section alone.
2. **Data.** Briefly describe the dataset: how many observations, what the variables are, and what the response variable is. Include one or two plots or summary statistics if they are informative. Do not paste the raw data.
3. **Methods.** State which model you fit and why it is appropriate for this type of response variable. One short paragraph is usually sufficient.
4. **Results.** Present your analysis: model selection, fitted model output, diagnostics, and interpretation. This is the longest section. Refer to numbered tables and figures rather than "the figure above".
5. **Conclusions.** Summarise the main findings in two or three sentences, and note the key limitations. What does the analysis tell us about the original question?
6. **Appendix.** Include all R code here, with brief comments. The appendix does not count towards the page limit. Code should not appear in the main body of the report.

Results: what to include for each analysis

- **Model selection:** explain which terms you kept or dropped and why. State the criterion used (e.g. AIC, likelihood ratio test) and your conclusion. Do not just list a sequence of models without explanation.
- **Goodness of fit:** report whether the model fits the data adequately. The appropriate check depends on the model family.
- **Diagnostics:** include suitable diagnostics and comment.
- **Interpretation:** do not just report coefficient estimates. Explain what they mean in the context of the problem.
- **Uncertainty:** report confidence intervals, not just point estimates.

Presentation

- Keep to the page limit.
- Use numbered figures and tables with informative captions. Refer to them by number in the text.
- Do not paste R output (or code) into the main body. Extract the relevant numbers and present them in a clean table or sentence.
- Use at least 11pt font and margins of at least 2cm on all sides.
- Proof-read your work. The spell-checker does not catch everything.
- Avoid informal language: "we got", "super small", "get rid of" are not appropriate in a statistical report.
- For individual reports, use "I" or passive constructions. For group reports, use "we".

Common mistakes

- Fitting a model without justifying why it is appropriate for the response variable.
- Dropping terms from a model without testing whether they are needed.
- Reporting a p-value without saying what conclusion it leads to.
- Using the word "significant" without having performed a hypothesis test.
- Reporting coefficients without interpreting them in context.
- Including plots that do not add anything to the analysis.
- Forgetting to check goodness of fit.
- Using causal language: write "is associated with" rather than "causes" or "leads to".

A note on R output

R output is evidence, not prose. Do not paste a `summary()` block into your report and expect the reader to interpret it. Extract the numbers you need, e.g. coefficients, standard errors, confidence intervals, AIC, and present them in a sentence or a clean table.